

Job Loss and Coping Strategies in Sub-Saharan Africa: Evidence from Custom Labor Force Surveys in Senegal

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Drawing on rich panel labor force surveys and custom worker behavior surveys in Senegal, we uncover how employment formality and income shape coping mechanisms following job loss in a highly informal labor market. Wage earners, particularly formal workers, support large networks of unemployed and inactive individuals, yet they navigate a volatile labor market with high employment transitions and low job security. Informal self-employed workers experience the highest turnover, facing frequent job changes and, on average, a 40% drop in household consumption upon job loss. Despite these challenges, most wage earners, especially poorer informal workers, lack strong coping mechanisms, struggling to pay outstanding loans and bills or access reliable financial support. While formal workers experience a larger proportional consumption loss due to their financial responsibilities, informal workers have fewer safety nets and limited access to credit, leaving them highly vulnerable. Covering a long enough period to capture labor market shifts, including the COVID-19 shock, these surveys provide rare insight into employment volatility and worker resilience and reveal a precarious dynamic: while formal workers face significant financial pressure from their dependents, informal workers, especially those who are poor, have the least ability to absorb income shocks. Recognizing these differences is critical for designing broad-based policies that strengthen employment protections, expand social safety nets, and help reduce poverty risks in regions characterized by high informality, high employment volatility, high reliance on social networks, and limited social protections.

Sub-Saharan Africa | Senegal | job loss | unemployment | coping strategy | informality

PLACEHOLDER FOR INTRODUCTION FOR NOW.

Poverty remains a persistent challenge in many regions of the world, particularly in Sub-Saharan Africa, where a large proportion of the population struggles to secure stable income sources and adequate living conditions. Despite notable efforts by governments, NGOs, and international development organizations, vulnerable households continue to grapple with economic uncertainties that threaten their livelihoods. One of the most pressing concerns in the region is the high prevalence of informal employment, which exists alongside a more structured formal labor sector. While the informal sector often offers flexibility and a degree of economic resilience in times of crisis, it also lacks the social protections and income stability typically associated with formal employment. These dual labor markets highlight the need to study how workers adapt to income shocks, what coping mechanisms they rely upon, and how policymakers can tailor interventions that address both immediate necessities and long-term development objectives.

This paper addresses these issues by analyzing data collected from Senegal, a country whose labor market dynamics reflect broader patterns observed across Sub-Saharan Africa. Through the combined use of the Employment Survey (ENES) and the Informal Sector Survey (ERISI), we gain valuable insights into the distribution of employment by region, the structural composition of the labor market, and how individuals and households respond to financial hardships. Focusing on a timeframe from 2015 to 2019, we capture an evolving economic landscape shaped by domestic policies, global market shifts, and local community support structures.

We build on (1) that presents stylized facts about the Senegalese labor market. There, we use nationally representative living standards and labour force surveys to document five key characteristics of the Senegalese labour market. Most workers are informal, with only a small fraction of workers and firms being formalized. Even within formal firms, there is a substantial presence of undeclared informal workers. There are pronounced income, consumption, and asset disparities across different employment statuses and, therefore, high potential

Significance Statement

Workers in Sub-Saharan Africa face severe challenges when they lose their jobs, yet responses differ sharply by employment type and income. Our research from Senegal shows that formal workers experience a larger proportional drop in consumption upon job loss, suggesting that more people depend on their earnings. Still, they have stronger coping mechanisms, such as access to credit and safety nets. In contrast, poor informal workers report a smaller share of consumption loss, likely due to limited discretionary spending, but they also have far fewer coping strategies—even within informal networks, leaving them highly vulnerable. These findings highlight the importance of strengthening labor protections and expanding social safety nets to mitigate economic shocks and support poverty reduction efforts.

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for consumption smoothing through social insurance. Even within formal employment settings, a majority of workers lack significant work-related benefits, highlighting the need for a more robust social safety net. Informal networks are a significant form of insurance for workers.

Our analysis integrates descriptive statistics and econometric modeling to map out the determinants of employment stability in various regions and industries. By highlighting the role of informal employment in buffering individuals against economic downturns, we also illustrate the potential for informal sector initiatives to act as stepping stones toward more secure livelihoods. At the same time, we underscore the inherent vulnerabilities that arise when social protections and legal frameworks are weak or absent—situations that can exacerbate the cycle of poverty for many households.

Ultimately, this research aims to inform policy discussions by offering a more granular view of how people navigate financial insecurity and employment transitions in impoverished contexts. In doing so, we hope to contribute practical recommendations for supporting micro-enterprises, strengthening community-based financial systems, and developing vocational training programs that align with the realities of the informal sector. The findings and policy implications herein serve as a call to action for national governments, international bodies, and civil society organizations committed to promoting inclusive growth and sustainable development in Sub-Saharan Africa.

Related Literature. CITE LITERATURE on JOB VULNERABILITY/INSECURITY, WAGE RIGIDITY, INFORMAL SECTOR, ETC. WAGE RIGIDITY (SUPREET KAUR JMP) LABOR MARKET FLOWS (KEVIN DONOVAN'S QJE PAPER) PAPERS ON JOB PROTECTION FROM OUR JME PAPER

A broad literature also examines how households cope with risk and the absence of formal insurance. In environments where formal unemployment or welfare benefits are lacking, families rely on informal arrangements such as mutual support networks, savings, and asset sales to smooth consumption. Classic studies in South Asia demonstrated that rural families use strategies like marrying into different villages or temporary migration to diversify risk across locations (2). In sub-Saharan Africa, informal credit and reciprocal transfers serve a similar insurance role: for instance, credit transactions in northern Nigeria effectively pool risk between borrowers and lenders when incomes fluctuate (3). At the same time, these informal insurance mechanisms are far from complete. Studies of African households experiencing major shocks find that consumption often drops significantly, indicating only partial insurance. In rural Burkina Faso, households facing a drought were able to smooth consumption very little, despite using grain stocks and livestock sales—pointing to serious limitations of informal risk-sharing and self-insurance (4). The inability of communities to fully insure against aggregate shocks has important consequences for poverty and vulnerability.

Migratory labor is another important coping mechanism highlighted in the literature. (2) showed how sending a family member to work elsewhere (for example, an urban area) can allow rural households to diversify income sources

and buffer agricultural shocks. More recent experimental evidence reinforces the value of migration in mitigating risk: (5) find that facilitating seasonal migration from famine-prone rural areas in Bangladesh led to higher earnings and improved food security for source households during the lean season. Though this study is in South Asia, seasonal migration is also a relevant strategy in parts of Africa where rainfall and agricultural calendars create analogous seasonal fluctuations in labor demand. In general, when local job opportunities falter, migration (temporary or permanent) can provide an alternative income stream.

Finally, researchers have evaluated the role of external interventions in relaxing the constraints faced by vulnerable workers. Providing access to credit or insurance can significantly change household behavior and outcomes. For example, a field experiment in Ghana showed that when farmers were given improved access to credit and rainfall insurance, they increased their investment in agricultural inputs, leading to higher output (6). This finding implies that in the status quo, fear of risk (like drought) and liquidity constraints hold back investment—a theme consistent with the broader narrative that vulnerability and lack of safety nets can stunt long-run economic progress. Likewise, studies of cash transfer programs illustrate the importance of informal sharing: (7) document that in Mexico's Progresa program, even households who were not direct beneficiaries experienced consumption gains because recipient households shared some of their transfer income with neighbors and relatives. Such spillover effects underscore that informal support networks remain active even when formal assistance is introduced, and they form a critical part of how communities respond to shocks and policy changes.

Results

Panel A and Panel B of Figure 1 track worker status for the same individuals in the EHCVM across 5 years. Panel A illustrates the five-year employment transitions across formal, informal, unemployed, and inactive categories, providing insight into labor market dynamics before and after the COVID-19 pandemic. The results highlight significant labor mobility across all employment statuses, particularly among informal workers who experience substantial shifts in employment status over time. Many individuals who were informally employed in 2018–2019 did not remain in the same category by 2021–2022, with some transitioning into unemployment, inactivity, or even formal employment. The high turnover in informal employment suggests the precarious nature of such jobs, where job stability is often low, and workers face greater exposure to economic shocks. Even among formal workers, there is notable movement, reflecting job losses and employment restructuring over the five-year period. Unemployed individuals and inactive populations also display substantial movement, with some re-entering the labor market, highlighting the fluidity of employment participation. The data captures both upward and downward mobility, indicating that employment transitions are not one-directional. It is particularly important to consider that this period includes the COVID-19 pandemic, which led to severe economic disruptions, disproportionately affecting informal workers and increasing job insecurity across all employment categories.

Panel B of Figure 1 extends this analysis by examining transitions across employment types (self-employed, employees, unemployed, inactive). The results suggest that self-employed individuals experienced particularly high volatility, with a large share shifting into unemployment or wage employment over the five-year period. This aligns with the challenges faced by self-employed individuals during economic downturns, where limited access to social protections and credit constraints can force them to exit self-employment. Conversely, employees—who are primarily wage earners—also exhibit notable transitions, with many moving into unemployment or informal self-employment, reflecting the fragility of job security, particularly in times of economic uncertainty. The large movement in employment status across all categories further reinforces the dynamic nature of the labor market, where even smaller groups of workers experience considerable change over time. The disruptive effects of COVID-19 are likely a major contributing factor to these employment shifts, with many workers being forced out of wage employment and into informal arrangements, unemployment, or inactivity.

Panel C of Figure 1 presents data on worker confidence in job security across formal and informal employment between 2017 and 2019, offering key insights into perceptions of labor market stability. A consistent trend emerges where formal workers report significantly higher confidence in keeping their jobs over the next twelve months compared to informal workers. In all years, a substantial majority of formal workers express confidence in job retention, with over 70% responding “Yes” to the question. This is in stark contrast to informal workers, where confidence is notably lower. In 2017, less than 45% of informal workers expressed confidence in job retention, while approximately 34% responded “No” and 21% responded “I don’t know”. Over time, job confidence among informal workers fluctuates but remains consistently lower than that of formal workers. By 2019, only 52% of informal workers express confidence in keeping their jobs, with a significant share (32%) still uncertain about their job security. Meanwhile, for formal workers, the proportion of those responding “Yes” remains relatively stable, around 70%, though there is a slight increase in uncertainty in 2019. The differences between formal and informal workers highlight the precarity of informal employment, where job stability is less certain, and workers face higher risks of displacement. The persistence of a sizable share of workers responding “I don’t know” also suggests that job insecurity remains a concern even for those who are currently employed, particularly among informal workers. Over the three-year period, while formal worker confidence remains high and relatively stable, informal workers continue to experience higher uncertainty and lower perceived job stability, reinforcing the structural vulnerabilities of informal employment in the labor market.

Figure 2 provides a visualization of the primary coping mechanisms used by different age groups (Young, Middle-Aged, and Old) in both urban and rural settings. The results highlight significant reliance on family support across all demographics, though its extent varies by age and location.

Among young individuals, family is the overwhelmingly dominant coping mechanism, with over 80% of urban youth and nearly 79% of rural youth relying on family for financial support. Savings play a secondary role, with 16% of urban

youth and 16% of rural youth reporting the ability to rely on personal savings. Other mechanisms, such as investment or pensions, are rarely used, with less than 4% of rural youth and just under 2% of urban youth indicating reliance on investments. This suggests that young people, particularly in urban areas, have few independent coping strategies and remain highly dependent on family networks.

Among middle-aged individuals, family remains the most relied-upon coping strategy but to a lesser extent than among younger individuals. Urban middle-aged individuals rely on family at a rate of 58%, while rural middle-aged individuals report a lower but still substantial rate of 54%. Unlike young individuals, savings play a larger role, accounting for 30% of urban middle-aged workers and 37% of their rural counterparts. Investments are used slightly more than in the younger group, particularly in rural areas, where 7% of middle-aged individuals rely on investments. Interestingly, middle-aged individuals are the only group where a small fraction report pensions as a coping strategy (0.2% in urban areas and 0.15% in rural areas), though at very low levels. This suggests that as individuals move into middle age, they diversify their financial safety nets beyond just family, but family remains their primary fallback.

Among older individuals, family reliance decreases further but remains the most frequently reported coping mechanism. Urban older individuals report a 49% reliance on family, while rural older individuals report 44%. Savings are particularly important for older individuals, with 42% of rural elderly and 30% of urban elderly depending on savings. Investment reliance increases with age, with 5% of urban elderly and 11% of rural elderly indicating investments as a coping strategy. Notably, pensions begin to play a small but significant role for older individuals, with 8% of urban elderly and 0.6% of rural elderly relying on pensions. This pattern suggests that while older individuals attempt to shift toward more stable financial resources such as savings and pensions, family remains the dominant safety net, particularly in rural settings where formal retirement mechanisms are less accessible.

Panel E of Figure 2 shows the share of workers who serve as a main financial support for their households, segmented by contract type (informal vs. formal employment) and income group (poor vs. rich). The findings highlight significant reliance on wage earners for household expenses, with particularly high levels of dependence among formal workers. The graph shows that formal workers, both poor and rich, are more likely to be the primary support of their household compared to informal workers. Among formal workers, the proportion of both poor and rich individuals who support their household exceeds 75%, with overlapping confidence intervals suggesting relatively similar reliance across income groups. This suggests that holding a formal job, regardless of income level, is associated with a higher likelihood that an individual is financially responsible for their family. This could be due to the relative stability and predictability of formal employment, making these workers more reliable sources of income for their households.

In contrast, among informal workers, there is a stark disparity between poor and rich individuals. Poor informal workers are significantly less likely to be the main financial support for their household, with their share closer to 40-50%, whereas rich informal workers report a much higher level of

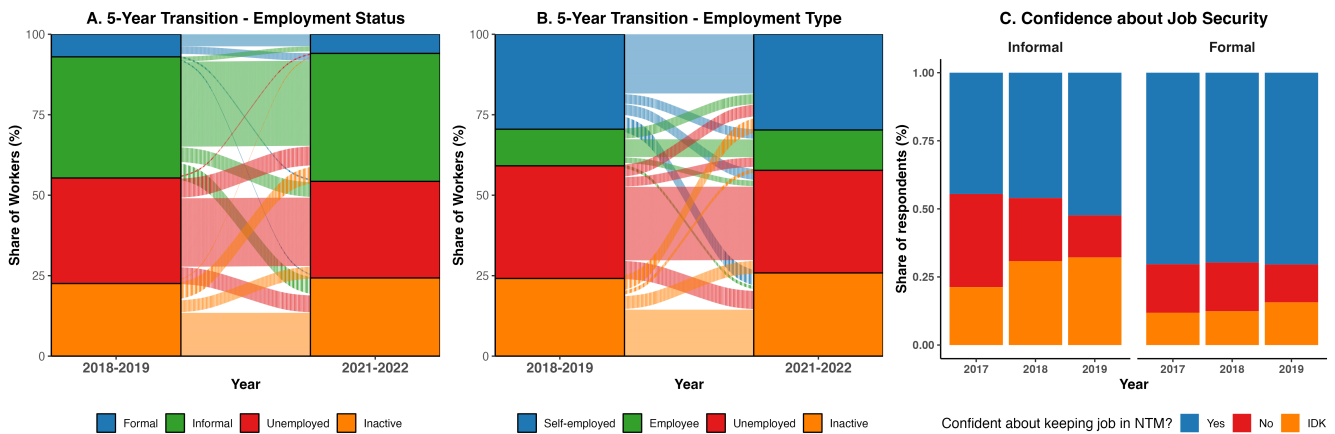


Fig. 1. High Transition Flows Across Employment Statuses and Low Worker Confidence about Job Security. Individuals are categorized as poor vs. rich workers and formal vs. informal workers. Poor (rich) workers are workers whose salary is below (above) median salary. Formal workers are workers with a written contract or with employment benefits. Panel A shows the change in household consumption as a share of current household consumption upon job loss. Panel B include both water and electricity bills. Panel C includes both formal and informal loans. Formal sources in Panel C refer to financial institutions officially registered. Informal sources refer to borrowings from informal lenders and social network. Savings in Panel C includes bank and mobile money savings, investments in real assets, and savings at home. Panel A and panel B use the EHCVM. Panel A tracks the employment status of respondents with different work status in the first round of the EHCVM (2018-2019) five years later in the second round of the EHCVM (2021-2022). Formal workers are wage workers with formal contracts. Informal workers are wage workers with no formal contracts. Unemployed individuals are people either looking for a job or not looking for a job due to involuntary reasons. Inactive people are people 15 years or older who are unemployed and not looking for a job for voluntary reasons. Panel B tracks the employment type of respondents in the first round of the EHCVM (2018-2019) five years later in the second round of the EHCVM (2021-2022). Employment type is as reported by the respondent. For Panel A and B, only respondents in the labor force (i.e. active people) are included in the analysis. Panel C uses the ENES data and displays the response to the question: "Are you confident about keeping your job in the next 12 months?", by formality type. "NTM" stands for next twelve months and "IDK" stands for I don't know. Both the EHCVM and the ENES data are described in the Materials and Methods section.

household dependence, approaching 70%. This difference suggests that informal workers with higher incomes may take on more financial responsibility within their families, despite working in a less stable employment arrangement. However, the lower level of reliance on poor informal workers could indicate that these individuals have less earning capacity and thus contribute less to household income, forcing families to rely on other sources of support, such as extended family or informal networks.

The high overall dependence on workers, particularly formal workers, underscores the critical role of wage earners in household financial stability. This suggests that job loss or income shocks affecting these workers could have widespread consequences, not just for individuals but for entire families. The results also highlight the economic burden placed on formal workers, who are often expected to support multiple dependents, which could contribute to financial stress even among higher-income individuals.

Panel A of Figure 3 shows the drop in Consumption as a share of total household consumption upon job loss. According to Panel A, both rich-formal and rich-informal workers exhibit a larger percentage drop in total household consumption when faced with job loss compared to poorer households. This seemingly paradoxical finding can be explained by the fact that wealthier households, who generally consume more overall, can afford—and therefore expect—to cut a greater share of their spending if income suddenly disappears. In contrast, poorer households have less discretionary or "non-essential" consumption to reduce in the first place, so their expected drop as a fraction of total consumption appears smaller. Where standard errors do not overlap—for instance, when comparing rich-formal to poor-informal—the difference in these consumption-cutting strategies is statistically significant. Despite this larger

percentage drop, having more resources overall (by virtue of both higher income and often more robust safety nets) still means that rich households are less likely to face dire hardships than the poor, even though the percentage cut in their spending is higher.

Panels B and C of Figure 3 relate to utility bills and loan repayment) and focus on more specific financial obligations—paying monthly utility bills and repaying outstanding loans. Overall, the ability to pay outstanding utility bills is low (around 40%) and the ability to pay back loans is even lower (less than 20%); and the difference across groups is not as stark as in panel A. The difference between rich-informal and poor-informal workers and between rich-informal and poor-formal is narrow, suggesting that formal contract status can partially offset a lower income level in terms of ability to pay back loans and outstanding bills. These findings reinforce that formal employment typically improves access to institutional protections (like severance or insurance), while higher income mitigates short-term cash flow disruptions.

Panels D and E of Figure 3 display access to formal and informal borrowing. They examine how likely workers are to borrow from formal institutions (banks, microfinance, etc.) versus informal networks (family, friends, or community groups) after job loss. The highest bar in Panel D belongs to poor-formal workers, whose standard errors do not overlap with those of poor-informal individuals, pointing to a significant difference in their access to official credit channels. Even poor-formal workers may rival or surpass rich-informal workers in Panel D, underscoring the distinct advantage conferred by a formal contract. In Panel E, although rich-formal respondents also lead in informal borrowing, the gap between them and other groups tends to be less stark, with some overlap in error bars. In other words, being formal consistently helps most with formal lending options,

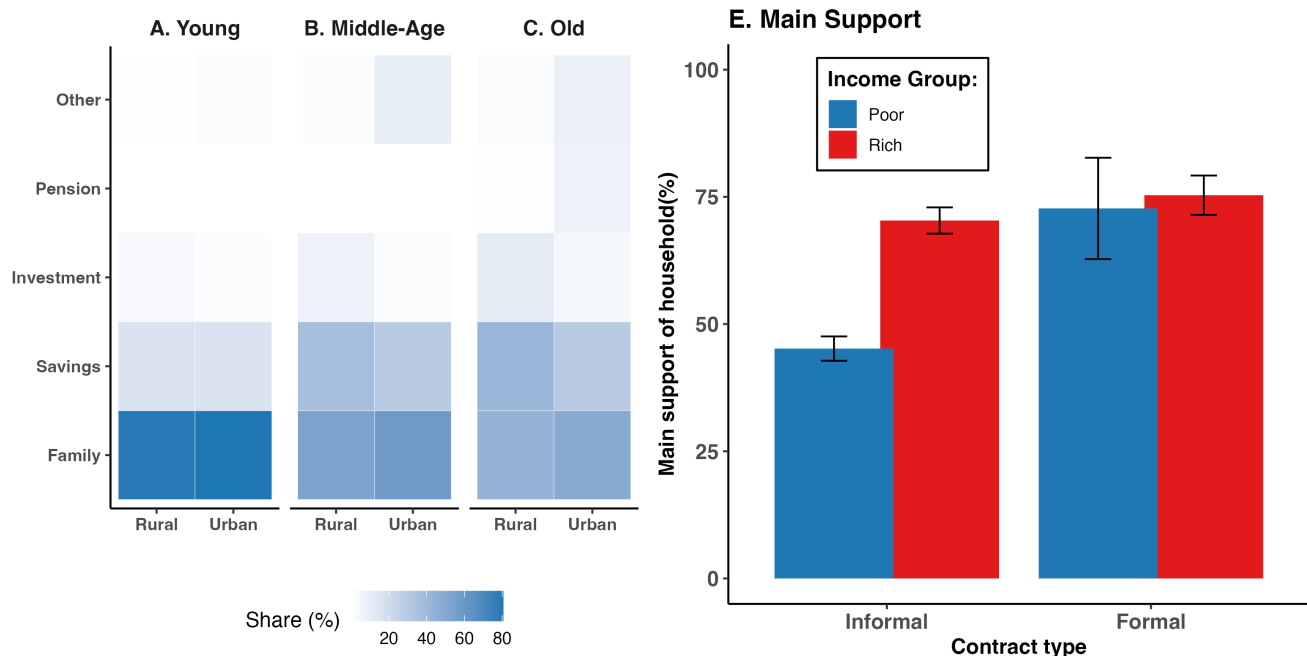


Fig. 2. High Dependence of Inactive and Unemployed Individuals on Family Wage Earners. Add description here.

while informal borrowing might depend more on social or familial networks that are not strictly tied to income level or employment status.

Panel F of Figure 3 displays the monthly Savings as a share of salary and shows the percentage of salary saved each month. Here, rich-formal households stand out with notably higher saving rates than poor-informal households—once again, with non-overlapping error bars suggesting that the difference is statistically significant. Rich-informal also save more, on average, than either poor group, although the advantage over poor-formal can be modest if their error bars overlap. Taken together, these six panels demonstrate that despite experiencing a larger percentage drop in consumption (Panel A), wealthier individuals—especially those with formal contracts—are better able to cope with unemployment through higher savings, stronger capacity to pay bills and loans, and greater access to both formal and informal credit. Meanwhile, poorer households, especially in the informal sector, face relatively smaller consumption cuts in percentage terms simply because they lack discretionary spending but also contend with constrained resources to survive prolonged joblessness, highlighting a significant vulnerability gap in times of economic distress.

Discussion

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The analysis shows a pronounced concentration of establishments in the capital city of Dakar, where 83% of businesses operate. Only 17% of establishments are located in other regions, highlighting a significant regional imbalance. This urban-centric distribution of economic opportunities has direct implications for rural populations, who often face higher employment precarity and limited access to formal job openings. Consequently, rural workers have fewer options for stable, regulated employment and are more susceptible

to economic shocks, underscoring the importance of region-specific interventions and infrastructure development.

Our findings further indicate that 39.1% of establishments employ fewer than five workers, illustrating the dominance of small-scale enterprises in Senegal's labor market. This prevalence of micro- and small enterprises corresponds with high levels of informality, reflecting a flexible yet precarious form of employment for many individuals. Indeed, the informal sector absorbs a considerable portion of the labor force, functioning as a primary buffer against income loss when formal job opportunities are scarce. While this dynamic highlights the sector's importance as a safety net, it also emphasizes the challenges in extending social protections, improving working conditions, and facilitating business growth within the informal economy.

In response to income shocks, a notable number of workers navigate between the formal and informal sectors to sustain their livelihoods. During economic downturns or personal financial crises, these workers rely on the informal sector's relative ease of entry to obtain temporary or supplementary income. Additionally, social safety nets, including government programs and community-based financial mechanisms such as tontines, serve as critical sources of support. Tontines, in particular, offer community-level savings and credit opportunities that can smooth consumption and mitigate the impact of sudden job loss or reduced earnings. By combining informal support systems with periodic formal employment, many households are able to maintain a semblance of financial stability, at least in the short term.

Our findings suggest several policy interventions to reduce employment vulnerability in impoverished regions, including enhancing labor market flexibility while promoting robust social protections, supporting micro-enterprises through improved access to finance and capacity-building programs, and expanding vocational training tailored to the needs

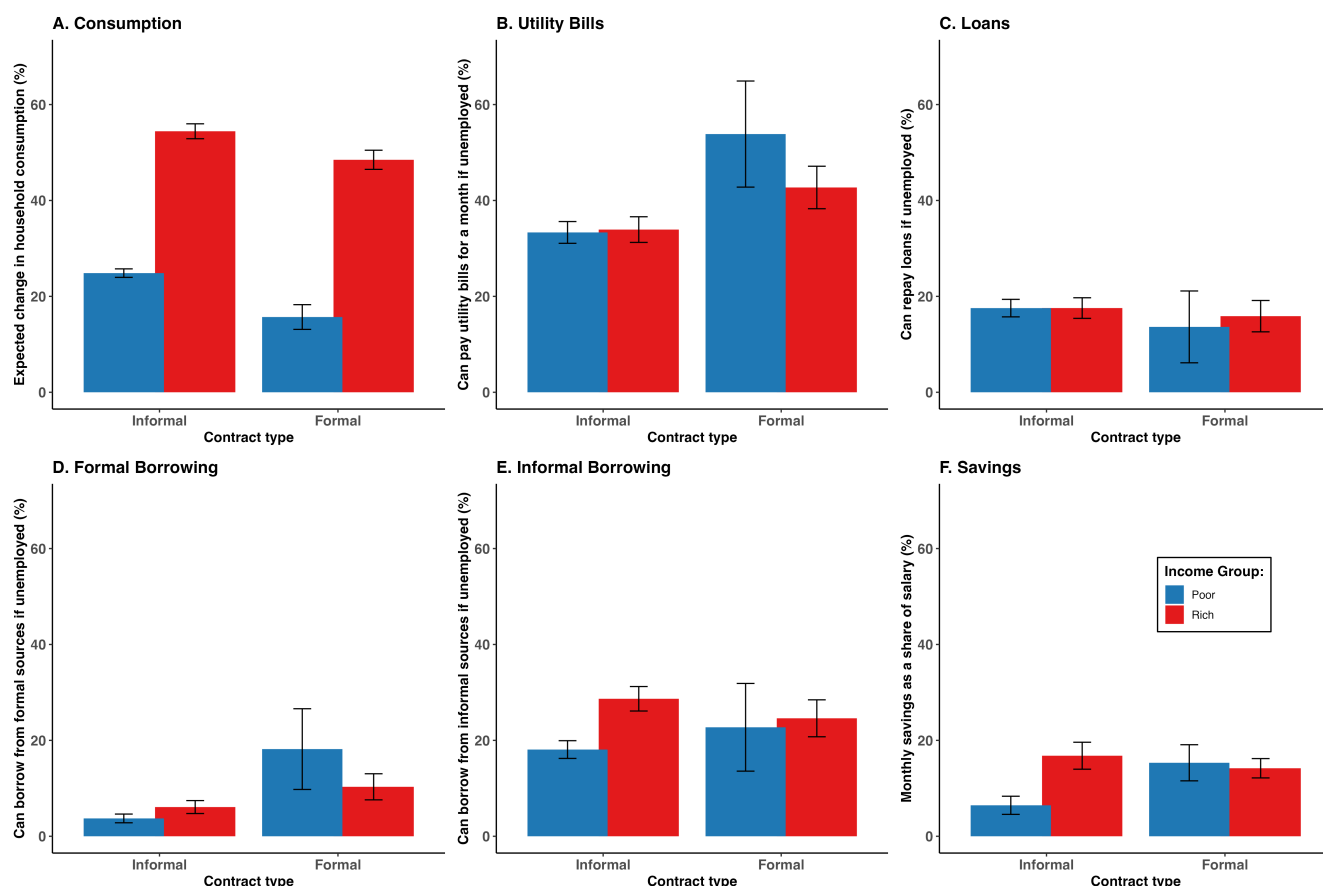


Fig. 3. Limited Coping Options upon Job Loss for Employed Individuals. Individuals are categorized as poor vs. rich workers and formal vs. informal workers. Poor (rich) workers are workers whose salary is below (above) median salary. Formal workers are workers with a written contract or with employment benefits. Panel A shows the change in household consumption as a share of current household consumption upon job loss. Panel B include both water and electricity bills. Panel C includes both formal and informal loans. Formal sources in Panel C refer to financial institutions officially registered. Informal sources refer to borrowings from informal lenders and social network. Savings in Panel C includes bank and mobile money savings, investments in real assets, and savings at home. All figures use the custom labor force survey described in the Materials and Methods section. Only respondents employed at the time of the survey are included in the analysis. Vertical lines show standard errors.

of the informal sector. These measures would not only address immediate economic challenges but also foster a more resilient labor market environment capable of weathering future shocks.

Materials and Methods

We use employment survey data spanning 2015–2019, focusing on: employment distribution by sector (formal/informal) and region, and household income shocks and response strategies, the role of informal employment in economic resilience. Data were analyzed using descriptive statistics and econometric models to understand the determinants of employment stability and adaptation to income shocks.

Materials. First, we conducted a custom labor force survey with a representative sample of the urban population in Senegal. This approach aligns with common practices for labor force surveys conducted in low-income countries, which primarily focus on urban areas. Thus, we abstract from the spillover effects of labor market policies on rural migration emphasized in the literature (8, 9). Furthermore, rural areas, where agricultural workers are typically found in Senegal, have dedicated government programs, such as agricultural

input subsidies, which could mitigate any potential effect of UI on rural–urban migration.*

The design of the custom survey follows a stratified random sampling approach. First, we define the population of the study as all active workers, which are individuals aged 15 or above, in Dakar, the capital of Senegal. Second, we use enumeration areas (EAs) as our primary sampling units (PSUs), as defined by the national statistical agency during the 2013 population census of Senegal. These EAs are distributed across the five districts in the region of Dakar: Dakar district, Guediawaye, Keur Massar, Pikine, and Rufisque. We randomly select 23 EAs from the set of 129 in Dakar. Third, within each selected EA, we randomly sample 15 households. The survey thus covers all individuals aged 15 and over within these selected households. In total, we survey 1,378 individuals across 345 households.

Second, we use longitudinal household expenditure surveys conducted in Senegal in 2018/2019 and in 2021/2022, the Enquête Harmonisée sur les Conditions de Vie des Ménages 2018–2019 (EHCVM) and the Enquête Harmonisée sur les Conditions de Vie des Ménages 2021–2022, to incorporate in

* See the discussion of agricultural workers and the reasons for their exclusion in the analysis in Section ??.

the analysis socioeconomic variables beyond those related to electricity that the datasets above provide. These surveys are very similar in spirit to the Living Standard Measurement Survey (LSMS) commonly conducted in several low-income countries. The data were collected by Agence Nationale de la Statistique et de la Démographie (ANSD) through a 2-stage sampling methodology. A total of 598 enumeration areas (EAs) were selected in the first stage, and 12 households were randomly selected in each EA in the second stage. The total survey sample size is 7,156 households, with 3,941 from urban areas and 3,215 from rural areas. The EHVCM is a rich dataset covering education, health, employment, nonemployment income, saving and credit, food consumption, food security, nonfood consumption, nonagricultural enterprises, housing, household assets, transfers, shocks and survival strategies, safety nets, agriculture, livestock, fishing, agricultural equipment, and relative poverty. Importantly, I have the exact centroids of the EAs where respondents live. I use the centroids to match the EHCVM with the rest of the datasets.

Third, we used the *Enquête Nationale sur l'Emploi au Senegal 2015-2019* (ENES). The ENES consists of 12 waves of quarterly, nationally representative labor force surveys from between 2015 and 2019. The survey uses a rotating panel of households. The ENES covers (i) demographic information on education, gender, age, and family structure;

(ii) information on employment status, contract structure, industry, occupation, earnings, working hours, formality type, tenure in the current job, and any changes in employment over the past three months; (iii) job search behavior with information on whether respondents engage in job search activities, the methods they employ in their job search, reasons for not actively seeking a job, and success in finding employment; (iv) consumption expenditures with information on the amount of money spent on food and beverages, utilities, and housing and any changes in these expenditures over the past few months; and (v) savings and borrowing with information on the methods used for saving and borrowing, the amount saved or borrowed, and whether the borrowing channels are formal or informal.

Fourth, we used the two waves of the *Enquête régionale intégrée sur l'emploi et le secteur informel*, which is representative survey of the informal sector.

Methods. WHAT ARE THE METHODS HERE. ANY REGRESSION WE CAN RUN?

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